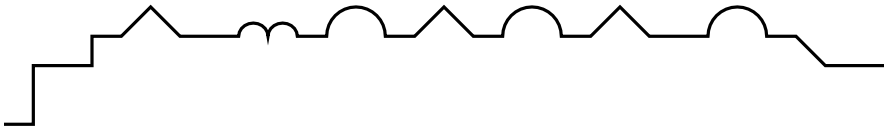
Math Kangaroo Practice

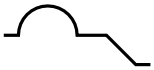
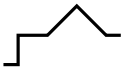
Grade 3-4 · 20 questions · harder · seed 67272

Part A: Each question is worth 3 points

1. Lena cut a strip of paper along this line.



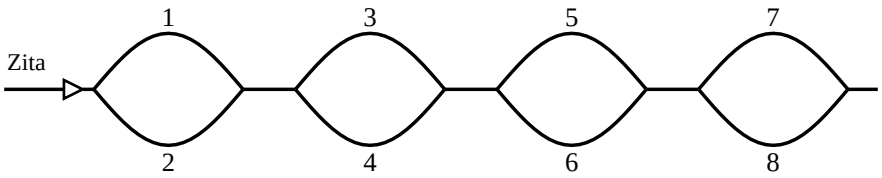
Which piece **cannot** come from the cut line?

A. ***B.***  ***C.*** ***D.***  ***E.***

2. Sara builds a straight fence from 17 pieces: posts and boards take turns, and the fence starts and ends with a post.
How many boards are there?

A. 9 ***B.*** 8 ***C.*** 7 ***D.*** 6 ***E.*** 5

3. Zita walks along the road from left to right and writes down every number on the way.

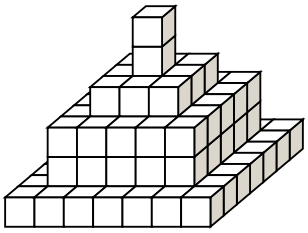


How many different groups of numbers can Zita write?

A. 18 ***B.*** 17 ***C.*** 16 ***D.*** 8 ***E.*** 4

Part B: Each question is worth 4 points

4. Robert built this pyramid out of identical cubes. Every layer is a full square, with no holes hidden inside.



How many cubes are in the pyramid?

A. 132 ***B.*** 123 ***C.*** 117 ***D.*** 110 ***E.*** 22

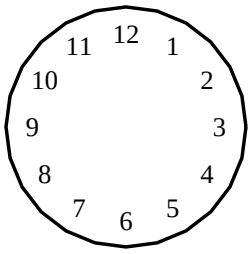
5. Dan has a rope 2 m long and cuts it into pieces that are each 20 cm long.
How many pieces does Dan get?

A. 1 ***B.*** 10 ***C.*** 11 ***D.*** 20 ***E.*** 100

6. Lily has the cards 1, 2, 6 and 9. Lily must use every card exactly once, with +, –, ×, ÷ and brackets.
Which number can Lily *not* get?

A. 40 ***B.*** 38 ***C.*** 35 ***D.*** 32 ***E.*** 28

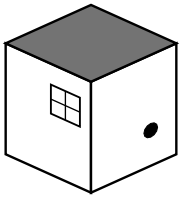
7. The face of a clock is cracked into 3 pieces. The sums of the numbers in the pieces are three consecutive numbers.



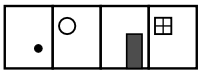
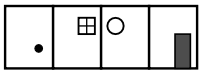
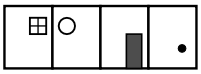
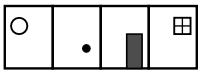
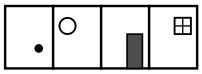
What is the largest of these sums?

A. 27 ***B.*** 28 ***C.*** 29 ***D.*** 30 ***E.*** 32

8. A strip of four walls is cut out and folded into this house, with the pictures on the outside. A flat roof is put on top.



Which strip was it?

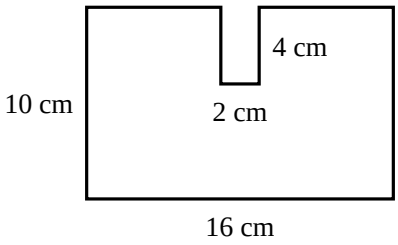
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

Part C: Each question is worth 5 points

9. Hugo paints a flag with three stripes: top, middle and bottom. Hugo has 6 colours (green, blue, white, red, yellow and orange). All three stripes must be different colours.
How many different flags can Hugo paint?

A. 120 ***B.*** 30 ***C.*** 20 ***D.*** 18 ***E.*** 15

10. A 16 cm by 10 cm rectangle of cardboard has a rectangular notch 2 cm wide and 4 cm deep cut out of the middle of its top side.



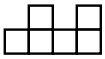
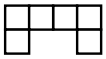
What is the perimeter of the new shape?

A. 58 cm ***B.*** 60 cm ***C.*** 64 cm ***D.*** 65 cm ***E.*** 72 cm

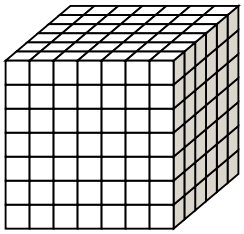
11. How many three-digit numbers have digits that add up to 8?

A. 8 ***B.*** 9 ***C.*** 36 ***D.*** 37 ***E.*** 72

12. Mia has two copies of the same piece. With which piece can she make a rectangle? Pieces may be turned and flipped over.

A. ***B.*** ***C.*** ***D.***  ***E.*** 

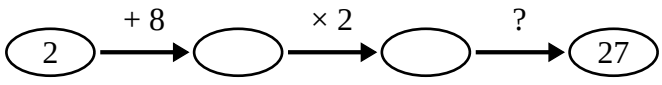
13. Mia builds a brick 7 cubes long, 6 cubes deep and 7 cubes tall from red and blue cubes. All the cubes you can see from the outside are red. The cubes just below them are **also** red, so on every side (top, bottom, front, back, left and right) there are two layers of red cubes. All the other cubes, in the middle, are blue.



How many blue cubes are there?

A. 48 ***B.*** 18 ***C.*** 16 ***D.*** 15 ***E.*** 6

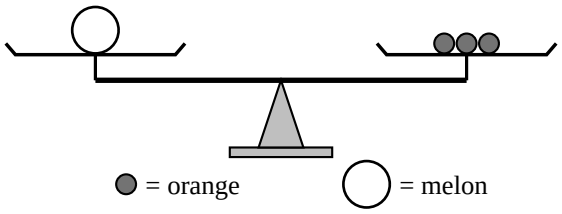
14. One arrow has lost its calculation. The others are done one after the other.



Which calculation belongs on the arrow with the ?

A. – 9 ***B.*** – 7 ***C.*** + 7 ***D.*** – 2 ***E.*** × 3

15. The balance shows that one melon weighs as much as 3 oranges. 1 melon and 4 oranges weigh 840 g altogether.



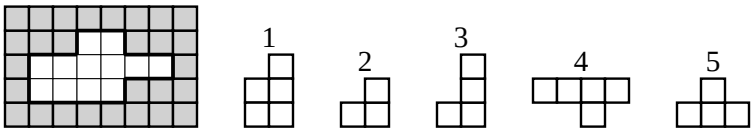
How much does one orange weigh?

A. 60 g ***B.*** 95 g ***C.*** 105 g ***D.*** 115 g ***E.*** 120 g

16. The odometer of a scooter shows 4589 km. Each digit is bigger than the digit before it. After how many more kilometres will this happen again?

A. 1 ***B.*** 79 ***C.*** 88 ***D.*** 89 ***E.*** 100

17. Which three of these pieces can be used to cover the empty area exactly? The pieces may be turned and flipped over.



A. 1+2+3 ***B.*** 1+2+4 ***C.*** 1+2+5 ***D.*** 1+3+4 ***E.*** 3+4+5

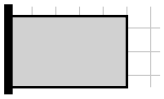
18. Leo has letter cards and wants to show which letters are used in the words CHOCOLATE and PENCIL. Leo puts down one card for each different letter used in either word.
How many cards does Leo put down?

A. 13 ***B.*** 10 ***C.*** 9 ***D.*** 8 ***E.*** 7

19. Exactly 483 digits are used to write all the house numbers in a street, starting from 1.
How many houses are there?

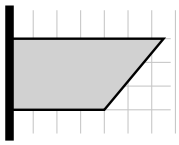
A. 237 ***B.*** 197 ***C.*** 196 ***D.*** 161 ***E.*** 98

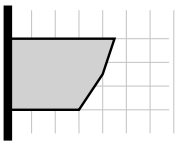
20. A rectangular flag flies on a pole in front of a wall, and the sun throws its shadow onto the wall. The wind turns and folds the flag, but the side sewn to the pole always stays on the pole. Tom draws the shadow at five different moments. Flat, the flag is 5 squares wide and 3 squares tall (see the picture). Four of the drawings are wrong.

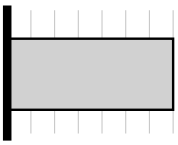


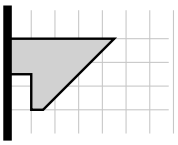
the flag, flat

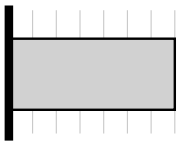
Which drawing can be right?

A. 

B. 

C. 

D. 

E. 

Answer Key

1.	**C**	**2.**	**B**	**3.**	**C**	**4.**	**D**	**5.**	**B**	**6.**	**B**
7.	**A**	**8.**	**E**	**9.**	**A**	**10.**	**B**	**11.**	**C**	**12.**	**D**
13.	**B**	**14.**	**C**	**15.**	**E**	**16.**	**D**	**17.**	**A**	**18.**	**B**
19.	**B**	**20.**	**B**								

Solutions

1. C — All the other pieces can be found in the line, but this one cannot: it has a round dome where the line has two small bumps. [2002 Q7 · cannot_belong]

2. B (8) — Post, board, post, ..., post: there is one more post than boards. 17 = 9 + 8, so 9 posts and 8 boards. [2003 Q17 · fence]

3. C (16) — The forks have 2, 2, 2, 2 branches; every choice can go with every other: 2 × 2 × 2 × 2 = 16. [2007 Q1 · how_many_routes]

4. D (110) — Count layer by layer, from the bottom: 7 × 7 + 5 × 5 × 2 + 3 × 3 + 1 × 1 × 2 = 110. [2002 Q20 · count_pyramid]

5. B (10) — 2 m = 200 cm. 200 ÷ 20 = 10. [2003 Q10 · pieces]

6. B (38) — 1 + 6 × 9 ÷ 2 = 28; (1 + 6 + 9) × 2 = 32; (6 – 2) × 9 – 1 = 35; (1 + 9) × (6 – 2) = 40. But with 1, 2, 6 and 9 nothing gives 38 (the closest are 37 = 1 + (6 – 2) × 9 and 40 = (1 + 9) × (6 – 2)). [2002 Q21 · which_cannot]

7. A (27) — All the numbers add up to 78. Three consecutive numbers that add up to 78: 25 + 26 + 27 = 78. The largest is 27. [2002 Q11 · largest_sum]

8. E — The left wall of the house must come right before the right wall on the strip (or the right wall is first and the left wall last), with each mark on the same side. Only strip E works. [2003 Q12 · which_strip]

9. A (120) — Top stripe: 6 choices, middle: 5, bottom: 4. 6 × 5 × 4 = 120. [2002 Q22 · stripes]

10. B (60 cm) — The bottom of the notch (2 cm) replaces the 2 cm taken out of the top edge, and the two sides of the notch add 2 × 4 cm. Perimeter = 2 × (16 + 10) + 2 × 4 = 60 cm. [2002 Q6 · notch]

11. C (36) — Count by the first digit: 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1 = 36. [2007 Q4 · count_with_sum]

12. D — Two copies have 12 squares, so the rectangle is 1 × 12 or 2 × 6 or 3 × 4. Only two copies of piece D fit together like that. [2007 Q8 · two_copies]

13. B (18) — Each length loses two red cubes at each end, so it drops by 4: 3 × 2 × 3 = 18. [2003 Q9 · thick_wall]

14. C (+ 7) — Before that arrow we have 20; working back from 27, after it we need 27. Only + 7 turns 20 into 27. [2007 Q5 · which_operation]

15. E (120 g) — Swap each melon for 3 oranges: 1 × 3 + 4 = 7 oranges weigh 840 g. One orange: 840 ÷ 7 = 120 g. [2002 Q9 · grams]

16. D (89) — Keep as many digits on the left as possible. The next such number is 4678: 4678 – 4589 = 89. [2002 Q24 · going_up]

17. A (1+2+3) — The empty area has 12 squares, so the three sizes must add up to 12. Only 1 + 2 + 3 also fit together in the right shape. [2003 Q18 · three_pieces]

18. B (10) — CHOCOLATE uses 7 different letters and PENCIL uses 6. C, E and L are in both, so count them once: 7 + 6 – 3 = 10. [2007 Q3 · letter_cards]

19. B (197) — 1-9 use 9 digits, 10-99 use 90 × 2 = 180, together 189. The other 483 – 189 = 294 digits are three-digit numbers: 294 ÷ 3 = 98, from 100 to 197. [2003 Q16 · three_digit_pages]

20. B — Check two things: the pole edge must be covered all the way, and nothing may reach more than 5 squares from the pole. Only picture B passes both checks. [2002 Q15 · only_one_right]